

Title:

House Price Prediction Using Linear Regression

Objective:

To build a machine learning model that predicts house prices.

Tools Used:

- Python
- Pandas
- Scikit-Learn
- Matplotlib

Methadology:

1. Loaded the dataset.
2. Selected important numerical features.
3. Split data into training and testing sets.
4. Trained a Linear Regression model.
5. Evaluated the model using MAE, MSE and R2 Score.

Results:

MAE : 210907.95

MSE : 98691835089.62

R2 Score : 0.0323

```
PS D:\python task 2> python -u "d:\python task 2\main.py"
```

```
Dataset Shape: (4600, 18)
```

	date	price	bedrooms	...	city	state	zip	count
0	2014-05-02 00:00:00	313000.0	3.0	...	Shoreline	WA	98133	
1	2014-05-02 00:00:00	2384000.0	5.0	...	Seattle	WA	98119	
2	2014-05-02 00:00:00	342000.0	3.0	...	Kent	WA	98042	
3	2014-05-02 00:00:00	420000.0	3.0	...	Bellevue	WA	98008	
4	2014-05-02 00:00:00	550000.0	4.0	...	Redmond	WA	98052	

```
[5 rows x 18 columns]
```

```
Model Performance
```

```
MAE : 210907.9548851089
```

```
MSE : 986918350899.619
```

```
R2 Score : 0.03228720631044091
```

```
Statistics:
```

	English	Science	Math
count	4.000000	4.000000	4.000000
mean	76.250000	83.750000	79.500000
std	8.995369	5.123475	18.770544
min	65.000000	78.000000	56.000000
25%	73.250000	80.250000	68.750000
50%	76.500000	84.000000	82.500000
75%	79.500000	87.500000	93.250000
max	87.000000	89.000000	97.000000

```
Average Math Marks: 79.5
```



